Mixed-strategy Nash equilibrium and the Folk Theorem

Consider a game between a security guard G and a thief L in an apartment building. The guard has two possible actions: watch, V, or sleep, D. The thief also has two possible actions: rob, R, or do nothing, N. Payoffs are ordered as (G, L):

	R	N
V	$(5, -10)$	$(-2, 0)$
D	$(-10, 10)$	$(0, 0)$

1. State which profiles survive iterative elimination of strictly dominated strategies
2. Find the Nash equilibrium or equilibria
3. Draw a graph with the probability of playing V on the horizontal axis and the probability of playing R on the vertical axis. Mark the Nash equilibria in the graph
4. Draw a graph with G's payoff on the horizontal axis and L's payoff on the vertical axis. Find the feasible payoff set and mark the expected utility of the mixed strategy equilibrium
5. State the Folk Theorem. Mark the payoffs that are attainable as subgame-perfect Nash equilibrium payoffs in the infinitely repeated game

Solution

	R	N
V	$(\underline{5}, -10)$	$(-2, \underline{0})$
D	$(-10, \underline{10})$	$(\underline{0}, 0)$

1. There are no strictly dominated strategies. For the guard, V is better when the thief plays R, but D is better when the thief plays N:

$$5 > -10 \qquad 0 > -2$$

For the thief, N is better when the guard plays V, but R is better when the guard plays D:

$$0 > -10 \qquad 10 > 0$$

Conclusion: no strategy is eliminated. Therefore, all strategy profiles survive iterative elimination of strictly dominated strategies:

$$(V, R) \qquad (V, N) \qquad (D, R) \qquad (D, N)$$

2. First, we look for pure-strategy Nash equilibria.

The guard's best responses are:

$$BR_G(R) = V \qquad BR_G(N) = D$$

The thief's best responses are:

$$BR_L(V) = N \qquad BR_L(D) = R$$

Thus, there is no pure-strategy Nash equilibrium.

Now let

$$x = \Pr(V) \qquad y = \Pr(R)$$

The guard is indifferent between V and D when

$$u_G(V) = 5y - 2(1-y) = 7y - 2$$

$$u_G(D) = -10y$$

Therefore,

$$7y - 2 = -10y \iff 17y = 2 \iff y = \frac{2}{17} \approx 0{,}118$$

The thief is indifferent between R and N when

$$u_L(R) = -10x + 10(1-x) = 10 - 20x$$

$$u_L(N) = 0$$

Therefore,

$$10 - 20x = 0 \iff x = \frac{1}{2} = 0{,}5$$

The mixed-strategy Nash equilibrium is

$$x^* = \frac{1}{2} \qquad y^* = \frac{2}{17}$$

The expected payoff of the guard is

$$u_G^* = -10\frac{2}{17} = -\frac{20}{17} \approx -1{,}176$$

The expected payoff of the thief is

$$u_L^* = 0$$

Conclusion: there is no pure-strategy Nash equilibrium. The unique Nash equilibrium is mixed:

$$\Pr(V) = \frac{1}{2} = 0{,}5 \qquad \Pr(R) = \frac{2}{17} \approx 0{,}118$$

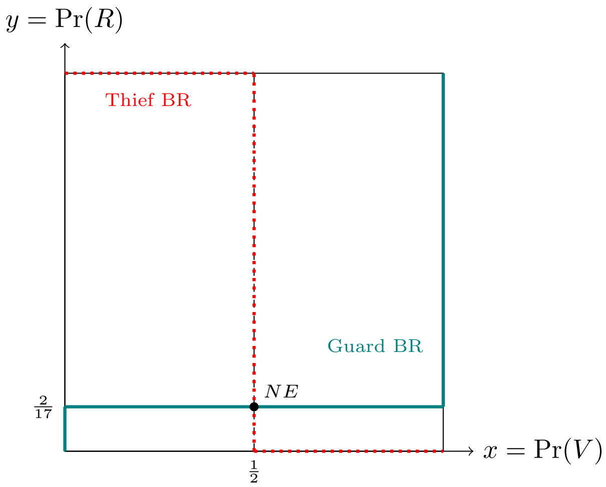
with expected payoffs

$$(u_G^*, u_L^*) = \left(-\frac{20}{17}, 0\right) \approx (-1{,}176, 0)$$

3. The best-response correspondences are:

$$BR_G(y) = \begin{cases} D & \text{if } y < \frac{2}{17} \\ \{V, D\} & \text{if } y = \frac{2}{17} \\ V & \text{if } y > \frac{2}{17} \end{cases}$$

$$BR_L(x) = \begin{cases} R & \text{if } x < \frac{1}{2} \\ \{R, N\} & \text{if } x = \frac{1}{2} \\ N & \text{if } x > \frac{1}{2} \end{cases}$$



Conclusion: the only intersection of the two best-response correspondences is

$$(x^*, y^*) = \left(\frac{1}{2}, \frac{2}{17}\right) \approx (0{,}5, 0{,}118)$$

4. The feasible payoff set is the convex hull of the payoff vectors generated by the stage game:

$$(5, -10) \qquad (-2, 0) \qquad (-10, 10) \qquad (0, 0)$$

The point $(-2, 0)$ lies inside the triangle formed by

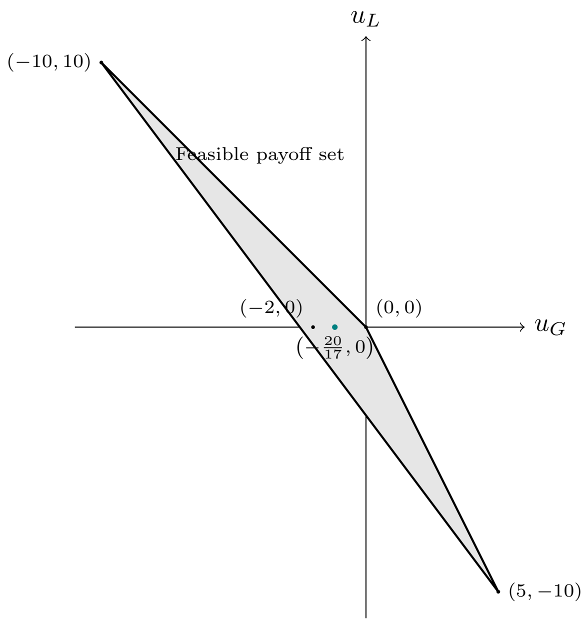
$$(5, -10) \qquad (-10, 10) \qquad (0, 0)$$

Therefore, the feasible payoff set is the triangle with vertices

$$(5, -10) \qquad (-10, 10) \qquad (0, 0)$$

The expected payoff of the mixed equilibrium found above is

$$(u_G^*, u_L^*) = \left(-\frac{20}{17}, 0\right) \approx (-1{,}176, 0)$$



Conclusion: the feasible payoff set is the convex hull of the stage-game payoff vectors, and the mixed-equilibrium payoff is

$$\left(-\frac{20}{17}, 0\right) \approx (-1{,}176, 0)$$

5. The Folk Theorem states that, in an infinitely repeated game, if players are sufficiently patient, then every feasible payoff vector that gives each player strictly more than his minmax payoff can be sustained as a subgame-perfect Nash equilibrium payoff.

   The minmax payoff of the guard is

   $$v_G = \min_y \max\{7y - 2, -10y\}$$

   The minimum occurs when

   $$7y - 2 = -10y \iff y = \frac{2}{17}$$

   Thus,

   $$v_G = -\frac{20}{17} \approx -1{,}176$$

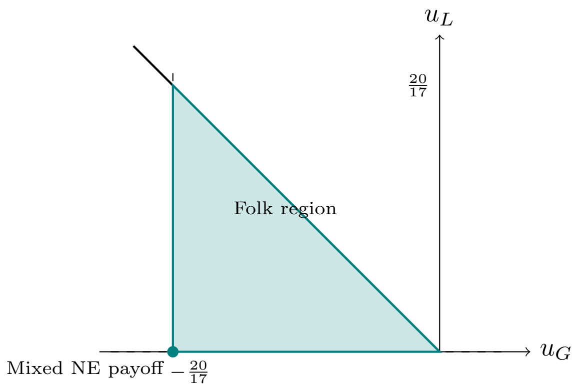
   The minmax payoff of the thief is

   $$v_L = \min_x \max\{10 - 20x, 0\} = 0$$

   Therefore, the strictly individually rational payoffs satisfy

   $$u_G > -\frac{20}{17} \qquad u_L > 0$$

   The payoffs sustained by the Folk Theorem are the feasible payoffs that satisfy these two inequalities.



Conclusion: for sufficiently high discount factors, the Folk Theorem sustains every feasible payoff vector satisfying

$$u_G > -\frac{20}{17} \approx -1{,}176 \qquad u_L > 0$$

The repeated mixed Nash equilibrium payoff

$$\left(-\frac{20}{17}, 0\right)$$

lies exactly at the minmax boundary